



RITAM MONDAL | 23IM30018

Indian Institute of Technology Kharagpur

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EDUCATION

Year	Degree/Exam	Institute
2028	B.Tech and M.Tech (DD), Industrial and system Engineering	Indian Institute of Technology Kharagpur
2022	Class XII, WBCHSE (87.8%)	Howrah Zilla School, Howrah
2020	Class X, WBBSE (91%)	Kona High School, Howrah

PROJECTS AND EXPERIENCE

Optimizing Delivery ETAs with Graph-Based Network Intelligence	April'26 – June'26	Summer Project'25 –IIT Guwahati (github)
- Engineered a directed weighted transportation graph with 1,657 nodes and 2,783 edges using 144K+ Delivery trip records. - Extracted graph analytics features including betweenness centrality, degree metrics, corridor delays, and node2vec embeddings. - Developed and benchmarked Graph-Enhanced XGBoost models, improving R² from 0.15 to 0.40 and reducing MAE from 16.26 to 14.53 minutes. - Built route intelligence framework for FTL vs Carting corridor analysis and operational decision support. - Deployed end-to-end Streamlit web application integrating Folium maps, OSRM API, ETA prediction, and delay-risk visualization.		
Web Based Attention Monitoring through eye tracking (IoT project)	Jan'26 – april'26	Mamata Jenamani, IIT Kharagpur (github)
- Integrated WebGazer.js frontend with backend APIs, enabling real-time gaze data collection and dashboard visualization. - Built an end-to-end gaze tracking system using MediaPipe, generating time-series data (screen coordinates, yaw, pitch) from webcam input. - Designed visual analytics (heatmaps, scanpaths, velocity plots) to interpret user attention patterns and behavior. - Deployed backend and enabled data export (CSV/JSON) with HTTPS-ready setup for live webcam-based applications. - Containerized the application using Docker, enabling consistent environment setup, dependency management, and easier deployment across systems.		
Optimizing Revenue Leakage & Profitability in the Hospitality Sector	May'25 – Jun'25	Summer Project'25 –IIT Guwahati (github)
- Built a full-stack analytics solution to identify causes of revenue loss in hotel bookings - Performed data preparation and EDA in Google Colab using Python (Pandas, Seaborn) - Developed Power BI dashboard with KPIs like ADR, RevPAR, Realisation %, Occupancy % - Simulated “late booking penalty” scenario — reduced revenue leakage by 49% - Provided strategic suggestions: weekday bundling, platform shift, non-refundable policy		
Rock Paper Scissors GUI Game with Audio-Visual Effects	June'25	Intern Project – Vault of Code (github)
- Developed an interactive Rock-Paper-Scissors game with Tkinter GUI and rich visuals for both player and bot. - Integrated image reactions (angry, scared, neutral) based on game outcome using PIL. - Added full audio experience using pygame: background music, button click sounds, and custom victory/loss effects. - Built game loop logic supporting 1–10 round battles, score tracking, and auto-ending with popup replay/exit options. - Final demo video showcased in GitHub README . - Demonstrated strong skills in GUI, Python threading, audio control, and creative user experience design.		
Portfolio Optimization in stock Market	Feb'25 - april'25	Avishek Sharma, IIT Kharagpur (github)
- Implemented Sharpe ratio optimization to construct an optimal stock portfolio - Used MATLAB and Python for data analysis, portfolio returns calculation, and risk assessment. - Processed real Excel data of 10 stocks spanning the past 2 years for analysis - Applied constrained optimization techniques to maximize risk-adjusted return - Determine the optimal allocation of stocks in a portfolio by deciding how much proportion of total investment should be assigned to each stock		
Forest Cover Type Prediction (Green Vision ML)	June'25 – July'25	PW skills (github)
<i>Machine Learning FastAPI AWS S3 Docker Full-Stack Deployment</i> - Predicted predominant forest cover type from cartographic features using USFS Region 2 dataset - Built ML pipeline with feature engineering, model training & evaluation (RandomForest, XGBoost) - Designed full-stack architecture with database integration and FastAPI web interface - Connected storage with AWS S3 bucket for dataset and model persistence - Containerized with Docker and deployed web app for interactive prediction		
Breast Cancer Detection Using Logistic Regression, SVC & GridSearchCV	May'25	self project (github)
<i>Machine Learning scikit-learn Python Data Analysis Model Evaluation</i> - Developed a binary classification model to predict malignant vs benign tumors using the Breast Cancer Wisconsin dataset - Built end-to-end ML pipeline with StandardScaler + GridSearchCV, tuning hyperparameters for LogisticRegression and SVC - Evaluated model using confusion matrix, ROC-AUC, precision/recall/accuracy trade-offs, and threshold optimization - Visualized key insights using 3D feature scatter plot, correlation heatmaps, and pie charts for class distribution		

SKILLS AND EXPERTISE

- Programming Languages:** Python, C++, C, MATLAB, R
- Machine Learning:** Regression, Classification, Ensemble Learning, XGBoost, Random Forest, SVM, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Cross Validation, ARIMA, SARIMA
- Deep Learning & Computer Vision:** Neural Networks, CNNs, Transfer Learning, Object Detection, Object Tracking, Image Classification, Image Segmentation, YOLO, MediaPipe
- Frameworks & Libraries:** Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, OpenCV, LangChain, Ollama
- Deployment & Tools:** FastAPI, Streamlit, Docker, AWS S3, Git, GitHub
- Data Analytics & Visualization:** EDA, Statistical Learning, Time Series Analysis, Power BI, Matplotlib, Seaborn
- Problem Solving:** 370+ LeetCode Problems, HackerRank 5★ Problem Solving

PARTICIPATION AND COURSES

Courses I learnt:

Academic: Probability and statistics | Supply chain management | statistical learning with application | Quality design and control | Simulation | Forecasting | Operations research | engineering economy | Management of Inventory systems | Algebra | Matlab-cplex-RStudio.

PW Skills - Data Science & Generative AI: Python, Machine Learning, Deep Learning, NLP, Databases, Generative AI, Feature Engineering, Model Development.

C++ with DSA: Arrays, Trees, BST, Graphs, Dynamic Programming, Segment Trees, Morris Traversal, Greedy Algorithms, and Competitive Programming.

CERTIFICATES

- C/C++ (PW skills) , Machine Learning (AlmaBetter),
- Python,Algorithmic Problem solving intermediate(Hacker Rank)
- Quant quest,Kshitij-IIT kgp'25 algorithmic trading certificate (by Unstop)

POSITIONS OF RESPONSIBILITY AND EXPERIENCE

Kshitij, Techno management fest of IIT kharagpur | Kshitij Campus Affiliate

Oct'23 - August'24

- Co-managed Kshitij 2024 with footfall of over 60000 students.
- Publicised the fest in the state of Jharkhand (Dhanbad, Ranchi, Deoghar) ,Bihar (patna), West Bengal (Kolkata, Hooghly) . Also make editors to publish the fest in their respective newspaper of Gano Shakti, Telegraph, Ananda bazar, Times of India

Student Mentor | at Physics Wallah

April'24 - August'24

- Mentored and guided JEE around 350 students for their preperation

AWARDS AND ACHIEVEMENTS

- Secured All India Rank of **5435** in **JEE Advanced 2023**(out of 1,90,000 candidates).
- Secured All India Rank of 20819 in JEE Mains 2023(out of 13,00,000 candidates).
- 5 stars in C program (Hackerrank), algorithmic problem solving (Leet Code 370+ questions)
- WBCHSE (class XII) - 87.8%
- WBBSE (class X) - 91%